

```
CREATE TABLE regions
( region_id      NUMBER
  CONSTRAINT region_id_nn NOT NULL
, region_name    VARCHAR2(25)
);
```

```
CREATE UNIQUE INDEX reg_id_pk
ON regions (region_id);
```

```
ALTER TABLE regions
ADD ( CONSTRAINT reg_id_pk
      PRIMARY KEY (region_id)
    );
```

```
CREATE TABLE countries
( country_id      CHAR(2)
  CONSTRAINT country_id_nn NOT NULL,
  country_name    VARCHAR2(40),
  region_id       NUMBER,
  CONSTRAINT country_c_id_pk
    PRIMARY KEY (country_id)
)
ORGANIZATION INDEX;
```

```
ALTER TABLE countries
ADD ( CONSTRAINT countr_reg_fk
      FOREIGN KEY (region_id)
      REFERENCES regions(region_id)
    );
```

```
CREATE TABLE locations
( location_id     NUMBER(4)
, street_address  VARCHAR2(40)
, postal_code     VARCHAR2(12)
, city           VARCHAR2(30)
  CONSTRAINT loc_city_nn NOT NULL
, state_province  VARCHAR2(25)
, country_id      CHAR(2)
);
```

```
CREATE UNIQUE INDEX loc_id_pk
ON locations (location_id);
```

```
ALTER TABLE locations
ADD ( CONSTRAINT loc_id_pk
```

```
    PRIMARY KEY (location_id)
, CONSTRAINT loc_c_id_fk
  FOREIGN KEY (country_id)
  REFERENCES countries(country_id)
);
```

```
CREATE SEQUENCE locations_seq
START WITH      3300
INCREMENT BY    100
MAXVALUE        9900
NOCACHE
NOCYCLE;
```

```
CREATE TABLE departments
( department_id    NUMBER(4)
, department_name  VARCHAR2(30)
  CONSTRAINT dept_name_nn  NOT NULL
, manager_id      NUMBER(6)
, location_id     NUMBER(4)
);
```

```
CREATE UNIQUE INDEX dept_id_pk
ON departments (department_id) ;
```

```
ALTER TABLE departments
ADD ( CONSTRAINT dept_id_pk
      PRIMARY KEY (department_id)
, CONSTRAINT dept_loc_fk
  FOREIGN KEY (location_id)
  REFERENCES locations (location_id)
);
```

```
CREATE SEQUENCE departments_seq
START WITH      280
INCREMENT BY    10
MAXVALUE        9990
NOCACHE
NOCYCLE;
```

```
CREATE TABLE jobs
( job_id          VARCHAR2(10)
, job_title       VARCHAR2(35)
  CONSTRAINT job_title_nn  NOT NULL
```

```
, min_salary    NUMBER(6)
, max_salary    NUMBER(6)
);
```

```
CREATE UNIQUE INDEX job_id_pk
ON jobs (job_id) ;
```

```
ALTER TABLE jobs
ADD ( CONSTRAINT job_id_pk
      PRIMARY KEY(job_id)
    );
```

```
CREATE TABLE employees
( employee_id    NUMBER(6)
, first_name     VARCHAR2(20)
, last_name      VARCHAR2(25)
  CONSTRAINT emp_last_name_nn NOT NULL
, email          VARCHAR2(25)
  CONSTRAINT emp_email_nn NOT NULL
, phone_number   VARCHAR2(20)
, hire_date      DATE
  CONSTRAINT emp_hire_date_nn NOT NULL
, job_id         VARCHAR2(10)
  CONSTRAINT emp_job_nn NOT NULL
, salary         NUMBER(8,2)
, commission_pct NUMBER(2,2)
, manager_id     NUMBER(6)
, department_id  NUMBER(4)
, CONSTRAINT emp_salary_min
  CHECK (salary > 0)
, CONSTRAINT emp_email_uk
  UNIQUE (email)
);
```

```
CREATE UNIQUE INDEX emp_emp_id_pk
ON employees (employee_id) ;
```

```
ALTER TABLE employees
ADD ( CONSTRAINT emp_emp_id_pk
      PRIMARY KEY (employee_id)
, CONSTRAINT emp_dept_fk
  FOREIGN KEY (department_id)
    REFERENCES departments
, CONSTRAINT emp_job_fk
  FOREIGN KEY (job_id)
    REFERENCES jobs (job_id)
```

```
, CONSTRAINT emp_manager_fk  
FOREIGN KEY (manager_id)  
REFERENCES employees  
);
```

```
ALTER TABLE departments  
ADD ( CONSTRAINT dept_mgr_fk  
FOREIGN KEY (manager_id)  
REFERENCES employees (employee_id)  
);
```

```
CREATE SEQUENCE employees_seq  
START WITH 207  
INCREMENT BY 1  
NOCACHE  
NOCYCLE;
```

```
CREATE TABLE job_history  
( employee_id NUMBER(6)  
CONSTRAINT jhist_employee_nn NOT NULL  
, start_date DATE  
CONSTRAINT jhist_start_date_nn NOT NULL  
, end_date DATE  
CONSTRAINT jhist_end_date_nn NOT NULL  
, job_id VARCHAR2(10)  
CONSTRAINT jhist_job_nn NOT NULL  
, department_id NUMBER(4)  
, CONSTRAINT jhist_date_interval  
CHECK (end_date > start_date)  
);
```

```
CREATE UNIQUE INDEX jhist_emp_id_st_date_pk  
ON job_history (employee_id, start_date) ;
```

```
ALTER TABLE job_history  
ADD ( CONSTRAINT jhist_emp_id_st_date_pk  
PRIMARY KEY (employee_id, start_date)  
, CONSTRAINT jhist_job_fk  
FOREIGN KEY (job_id)  
REFERENCES jobs  
, CONSTRAINT jhist_emp_fk  
FOREIGN KEY (employee_id)  
REFERENCES employees  
, CONSTRAINT jhist_dept_fk  
FOREIGN KEY (department_id)
```

REFERENCES departments

);

CREATE OR REPLACE VIEW emp_details_view

(employee_id,
job_id,
manager_id,
department_id,
location_id,
country_id,
first_name,
last_name,
salary,
commission_pct,
department_name,
job_title,
city,
state_province,
country_name,
region_name)

AS SELECT

e.employee_id,
e.job_id,
e.manager_id,
e.department_id,
d.location_id,
l.country_id,
e.first_name,
e.last_name,
e.salary,
e.commission_pct,
d.department_name,
j.job_title,
l.city,
l.state_province,
c.country_name,
r.region_name

FROM

employees e,
departments d,
jobs j,
locations l,
countries c,
regions r

WHERE e.department_id = d.department_id

AND d.location_id = l.location_id

AND l.country_id = c.country_id

```
    AND c.region_id = r.region_id  
    AND j.job_id = e.job_id  
WITH READ ONLY;
```

```
COMMIT;
```

```
INSERT INTO regions VALUES  
    ( 1  
      , 'Europe'  
    );
```

```
INSERT INTO regions VALUES  
    ( 2  
      , 'Americas'  
    );
```

```
INSERT INTO regions VALUES  
    ( 3  
      , 'Asia'  
    );
```

```
INSERT INTO regions VALUES  
    ( 4  
      , 'Middle East and Africa'  
    );
```

```
INSERT INTO countries VALUES  
    ( 'IT'  
      , 'Italy'  
      , 1  
    );
```

```
INSERT INTO countries VALUES  
    ( 'JP'  
      , 'Japan'  
      , 3  
    );
```

```
INSERT INTO countries VALUES  
    ( 'US'  
      , 'United States of America'  
      , 2  
    );
```

```
INSERT INTO countries VALUES  
    ( 'CA'  
      , 'Canada'
```

```
, 2  
);
```

```
INSERT INTO countries VALUES  
  ( 'CN'  
    , 'China'  
    , 3  
  );
```

```
INSERT INTO countries VALUES  
  ( 'IN'  
    , 'India'  
    , 3  
  );
```

```
INSERT INTO countries VALUES  
  ( 'AU'  
    , 'Australia'  
    , 3  
  );
```

```
INSERT INTO countries VALUES  
  ( 'ZW'  
    , 'Zimbabwe'  
    , 4  
  );
```

```
INSERT INTO countries VALUES  
  ( 'SG'  
    , 'Singapore'  
    , 3  
  );
```

```
INSERT INTO countries VALUES  
  ( 'UK'  
    , 'United Kingdom'  
    , 1  
  );
```

```
INSERT INTO countries VALUES  
  ( 'FR'  
    , 'France'  
    , 1  
  );
```

```
INSERT INTO countries VALUES  
  ( 'DE'  
    , 'Germany'
```

```
, 1  
);
```

```
INSERT INTO countries VALUES  
  ( 'ZM'  
    , 'Zambia'  
    , 4  
  );
```

```
INSERT INTO countries VALUES  
  ( 'EG'  
    , 'Egypt'  
    , 4  
  );
```

```
INSERT INTO countries VALUES  
  ( 'BR'  
    , 'Brazil'  
    , 2  
  );
```

```
INSERT INTO countries VALUES  
  ( 'CH'  
    , 'Switzerland'  
    , 1  
  );
```

```
INSERT INTO countries VALUES  
  ( 'NL'  
    , 'Netherlands'  
    , 1  
  );
```

```
INSERT INTO countries VALUES  
  ( 'MX'  
    , 'Mexico'  
    , 2  
  );
```

```
INSERT INTO countries VALUES  
  ( 'KW'  
    , 'Kuwait'  
    , 4  
  );
```

```
INSERT INTO countries VALUES  
  ( 'IL'  
    , 'Israel'
```



```
, 4  
);
```

```
INSERT INTO countries VALUES  
  ( 'DK'  
    , 'Denmark'  
    , 1  
  );
```

```
INSERT INTO countries VALUES  
  ( 'HK'  
    , 'HongKong'  
    , 3  
  );
```

```
INSERT INTO countries VALUES  
  ( 'NG'  
    , 'Nigeria'  
    , 4  
  );
```

```
INSERT INTO countries VALUES  
  ( 'AR'  
    , 'Argentina'  
    , 2  
  );
```

```
INSERT INTO countries VALUES  
  ( 'BE'  
    , 'Belgium'  
    , 1  
  );
```

```
INSERT INTO locations VALUES  
  ( 1000  
    , '1297 Via Cola di Rie'  
    , '00989'  
    , 'Roma'  
    , NULL  
    , 'IT'  
  );
```

```
INSERT INTO locations VALUES  
  ( 1100  
    , '93091 Calle della Testa'  
    , '10934'  
    , 'Venice'
```

```
, NULL  
, 'IT'  
);
```

INSERT INTO locations VALUES

```
( 1200  
, '2017 Shinjuku-ku'  
, '1689'  
, 'Tokyo'  
, 'Tokyo Prefecture'  
, 'JP'  
);
```

INSERT INTO locations VALUES

```
( 1300  
, '9450 Kamiya-cho'  
, '6823'  
, 'Hiroshima'  
, NULL  
, 'JP'  
);
```

INSERT INTO locations VALUES

```
( 1400  
, '2014 Jabberwocky Rd'  
, '26192'  
, 'Southlake'  
, 'Texas'  
, 'US'  
);
```

INSERT INTO locations VALUES

```
( 1500  
, '2011 Interiors Blvd'  
, '99236'  
, 'South San Francisco'  
, 'California'  
, 'US'  
);
```

INSERT INTO locations VALUES

```
( 1600  
, '2007 Zagora St'  
, '50090'  
, 'South Brunswick'  
, 'New Jersey'  
, 'US'  
);
```

INSERT INTO locations VALUES

```
( 1700
, '2004 Charade Rd'
, '98199'
, 'Seattle'
, 'Washington'
, 'US'
);
```

INSERT INTO locations VALUES

```
( 1800
, '147 Spadina Ave'
, 'M5V 2L7'
, 'Toronto'
, 'Ontario'
, 'CA'
);
```

INSERT INTO locations VALUES

```
( 1900
, '6092 Boxwood St'
, 'YSW 9T2'
, 'Whitehorse'
, 'Yukon'
, 'CA'
);
```

INSERT INTO locations VALUES

```
( 2000
, '40-5-12 Laogianggen'
, '190518'
, 'Beijing'
, NULL
, 'CN'
);
```

INSERT INTO locations VALUES

```
( 2100
, '1298 Vileparle (E)'
, '490231'
, 'Bombay'
, 'Maharashtra'
, 'IN'
);
```

INSERT INTO locations VALUES

```
( 2200
```

```
, '12-98 Victoria Street'  
, '2901'  
, 'Sydney'  
, 'New South Wales'  
, 'AU'  
);
```

INSERT INTO locations VALUES

```
( 2300  
, '198 Clementi North'  
, '540198'  
, 'Singapore'  
, NULL  
, 'SG'  
);
```

INSERT INTO locations VALUES

```
( 2400  
, '8204 Arthur St'  
, NULL  
, 'London'  
, NULL  
, 'UK'  
);
```

INSERT INTO locations VALUES

```
( 2500  
, 'Magdalen Centre, The Oxford Science Park'  
, 'OX9 9ZB'  
, 'Oxford'  
, 'Oxford'  
, 'UK'  
);
```

INSERT INTO locations VALUES

```
( 2600  
, '9702 Chester Road'  
, '09629850293'  
, 'Stretford'  
, 'Manchester'  
, 'UK'  
);
```

INSERT INTO locations VALUES

```
( 2700  
, 'Schwanthalerstr. 7031'  
, '80925'  
, 'Munich'
```

```
, 'Bavaria'  
, 'DE'  
);
```

```
INSERT INTO locations VALUES  
  ( 2800  
    , 'Rua Frei Caneca 1360 '  
    , '01307-002'  
    , 'Sao Paulo'  
    , 'Sao Paulo'  
    , 'BR'  
  );
```

```
INSERT INTO locations VALUES  
  ( 2900  
    , '20 Rue des Corps-Saints'  
    , '1730'  
    , 'Geneva'  
    , 'Geneve'  
    , 'CH'  
  );
```

```
INSERT INTO locations VALUES  
  ( 3000  
    , 'Murtenstrasse 921'  
    , '3095'  
    , 'Bern'  
    , 'BE'  
    , 'CH'  
  );
```

```
INSERT INTO locations VALUES  
  ( 3100  
    , 'Pieter Breughelstraat 837'  
    , '3029SK'  
    , 'Utrecht'  
    , 'Utrecht'  
    , 'NL'  
  );
```

```
INSERT INTO locations VALUES  
  ( 3200  
    , 'Mariano Escobedo 9991'  
    , '11932'  
    , 'Mexico City'  
    , 'Distrito Federal,'  
    , 'MX'  
  );
```

```
ALTER TABLE departments
  DISABLE CONSTRAINT dept_mgr_fk;
```

```
INSERT INTO departments VALUES
  ( 10
    , 'Administration'
    , 200
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 20
    , 'Marketing'
    , 201
    , 1800
  );
```

```
INSERT INTO departments VALUES
  ( 30
    , 'Purchasing'
    , 114
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 40
    , 'Human Resources'
    , 203
    , 2400
  );
```

```
INSERT INTO departments VALUES
  ( 50
    , 'Shipping'
    , 121
    , 1500
  );
```

```
INSERT INTO departments VALUES
  ( 60
    , 'IT'
    , 103
    , 1400
  );
```

```
INSERT INTO departments VALUES
```

```
( 70  
  , 'Public Relations'  
  , 204  
  , 2700  
);
```

INSERT INTO departments VALUES

```
( 80  
  , 'Sales'  
  , 145  
  , 2500  
);
```

INSERT INTO departments VALUES

```
( 90  
  , 'Executive'  
  , 100  
  , 1700  
);
```

INSERT INTO departments VALUES

```
( 100  
  , 'Finance'  
  , 108  
  , 1700  
);
```

INSERT INTO departments VALUES

```
( 110  
  , 'Accounting'  
  , 205  
  , 1700  
);
```

INSERT INTO departments VALUES

```
( 120  
  , 'Treasury'  
  , NULL  
  , 1700  
);
```

INSERT INTO departments VALUES

```
( 130  
  , 'Corporate Tax'  
  , NULL  
  , 1700  
);
```

```
INSERT INTO departments VALUES
    ( 140
    , 'Control And Credit'
    , NULL
    , 1700
    );
```

```
INSERT INTO departments VALUES
    ( 150
    , 'Shareholder Services'
    , NULL
    , 1700
    );
```

```
INSERT INTO departments VALUES
    ( 160
    , 'Benefits'
    , NULL
    , 1700
    );
```

```
INSERT INTO departments VALUES
    ( 170
    , 'Manufacturing'
    , NULL
    , 1700
    );
```

```
INSERT INTO departments VALUES
    ( 180
    , 'Construction'
    , NULL
    , 1700
    );
```

```
INSERT INTO departments VALUES
    ( 190
    , 'Contracting'
    , NULL
    , 1700
    );
```

```
INSERT INTO departments VALUES
    ( 200
    , 'Operations'
    , NULL
    , 1700
    );
```



```
INSERT INTO departments VALUES
  ( 210
    , 'IT Support'
    , NULL
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 220
    , 'NOC'
    , NULL
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 230
    , 'IT Helpdesk'
    , NULL
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 240
    , 'Government Sales'
    , NULL
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 250
    , 'Retail Sales'
    , NULL
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 260
    , 'Recruiting'
    , NULL
    , 1700
  );
```

```
INSERT INTO departments VALUES
  ( 270
    , 'Payroll'
    , NULL
    , 1700
  );
```

```
);  
INSERT INTO jobs VALUES  
  ( 'AD_PRES'  
    , 'President'  
    , 20000  
    , 40000  
    );  
INSERT INTO jobs VALUES  
  ( 'AD_VP'  
    , 'Administration Vice President'  
    , 15000  
    , 30000  
    );  
  
INSERT INTO jobs VALUES  
  ( 'AD_ASST'  
    , 'Administration Assistant'  
    , 3000  
    , 6000  
    );  
  
INSERT INTO jobs VALUES  
  ( 'FI_MGR'  
    , 'Finance Manager'  
    , 8200  
    , 16000  
    );  
  
INSERT INTO jobs VALUES  
  ( 'FI_ACCOUNT'  
    , 'Accountant'  
    , 4200  
    , 9000  
    );
```

```
INSERT INTO jobs VALUES
    ( 'AC_MGR'
    , 'Accounting Manager'
    , 8200
    , 16000
    );
```

```
INSERT INTO jobs VALUES
    ( 'AC_ACCOUNT'
    , 'Public Accountant'
    , 4200
    , 9000
    );
```

```
INSERT INTO jobs VALUES
    ( 'SA_MAN'
    , 'Sales Manager'
    , 10000
    , 20000
    );
```

```
INSERT INTO jobs VALUES
    ( 'SA_REP'
    , 'Sales Representative'
    , 6000
    , 12000
    );
```

```
INSERT INTO jobs VALUES
    ( 'PU_MAN'
    , 'Purchasing Manager'
    , 8000
    , 15000
    );
```

```
INSERT INTO jobs VALUES
  ( 'PU_CLERK'
  , 'Purchasing Clerk'
  , 2500
  , 5500
  );
```

```
INSERT INTO jobs VALUES
  ( 'ST_MAN'
  , 'Stock Manager'
  , 5500
  , 8500
  );
```

```
INSERT INTO jobs VALUES
  ( 'ST_CLERK'
  , 'Stock Clerk'
  , 2000
  , 5000
  );
```

```
INSERT INTO jobs VALUES
  ( 'SH_CLERK'
  , 'Shipping Clerk'
  , 2500
  , 5500
  );
```

```
INSERT INTO jobs VALUES
  ( 'IT_PROG'
  , 'Programmer'
  , 4000
  , 10000
  );
```

```
INSERT INTO jobs VALUES
( 'MK_MAN'
, 'Marketing Manager'
, 9000
, 15000
);
```

```
INSERT INTO jobs VALUES
( 'MK_REP'
, 'Marketing Representative'
, 4000
, 9000
);
```

```
INSERT INTO jobs VALUES
( 'HR_REP'
, 'Human Resources Representative'
, 4000
, 9000
);
```

```
INSERT INTO jobs VALUES
( 'PR_REP'
, 'Public Relations Representative'
, 4500
, 10500
);
```

```
COMMIT;
```